

MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in AI Security, Security Operations, Threat Hunting, Cloud Security, and Intelligent Cyber Defense. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on AI-driven cyber defense, intelligent threat detection, adaptive deception, and autonomous security architectures. Hands-on knowledge of Microsoft Sentinel, Kusto Query Language (KQL), Microsoft Defender Suite, Azure security services, MITRE ATT&CK, and AI/LLM security. Seeking opportunities in Cybersecurity Engineering, AI Security, Security Operations, and Cloud Security.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Security Operations: Microsoft Sentinel, SIEM Operations, Security Monitoring, Threat Hunting, Incident Response, Detection Engineering, IOC Analysis, SOAR Fundamentals, Threat Intelligence

Cloud Security: Microsoft Defender for Cloud, Microsoft Defender for Endpoint, Microsoft Defender for Containers, Microsoft Defender for Identity, Azure Monitor, Azure Key Vault, Azure Kubernetes Service (AKS), Identity and Access Management (IAM), Zero Trust Architecture

AI Security: AI-Driven Cyber Defense, Adversarial AI, Prompt Injection, Model Inversion, Membership Inference, AI API Security, OWASP LLM Top 10, AI Supply Chain Security

Threat Intelligence: MITRE ATT&CK Framework, Threat Hunting, IOC Analysis, MISP, Malware Analysis Fundamentals, Threat Modeling

Programming: Python, Java, JavaScript, SQL, Bash

Security Automation: Kusto Query Language (KQL), Logic Apps, Azure Functions, Python Automation

Application Security: OWASP Top 10, API Security, Secure Coding, Authentication, Authorization

Security Tools: Wireshark, Burp Suite, OWASP ZAP, Nmap, Sysmon, YARA, Sigma Rules

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP)

Databases: MySQL, MongoDB

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL supporting enterprise security operations and analytics.
- Implemented authentication, authorization, secure API communication, database security controls, and secure application workflows.
- Developed scalable backend services supporting security analytics, operational monitoring, intelligent decision support, and structured log processing.
- Engineered maintainable backend architectures emphasizing secure software engineering, cloud deployment readiness, enterprise reliability, and security-focused application development.

Data Science Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven intelligent analytics framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed an intelligent AI security architecture integrating AI-driven threat detection, threat hunting, adaptive deception, distributed threat intelligence, autonomous incident response, and resilient cloud security principles.
- Developed machine learning models integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honey-pots, reinforcement learning, IOC-driven analytics, and AI-assisted security operations.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise security scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed an intelligent secure platform integrating AI-assisted governance, authentication, trust scoring, fraud detection, secure transaction validation, cloud-ready security principles, and enterprise risk mitigation.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing machine learning-assisted validation, AES-256 encryption, SHA-256 hashing, and intelligent transaction analytics.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing an intelligent data-driven framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching feature engineering, intelligent optimization, semantic data representation, post-quantum cryptography, and privacy-preserving data analytics.
- Developing intelligent cryptographic security mechanisms supporting adaptive cyber defense, resilient trust management, secure enterprise architectures, and AI-enabled security systems.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Google Cloud Computing – NPTEL
- IBM Front-End Developer Specialization – IBM
- Systems and Usable Security (Elite) – NPTEL
- Ethical Hacker – Cisco Networking Academy
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO - LEOS).
- First Author of internationally recognized research focused on AI-driven cybersecurity, intelligent distributed systems, autonomous cyber defense, and secure software architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.